Joseph Farkas

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Education

Georgia Institute of Technology — Atlanta, GA

Bachelor of Science in Computer Science, Devices and Intelligence — *Expected Dec 2027*

Minors in Robotics (Aerospace Systems) and German; Concert Orchestra Violinist; BS/MS Applicant — *GPA: 4.00*

- Relevant Coursework: Data Structures, Algorithm Design and Analysis, Robotics and Perception

Professional Experience

Autonomous Systems Lead, Software Mentor — Oct 2021 – Present

North Gwinnett Robotics — *Suwanee, GA*

- Mentoring ~15 high school students in sensor fusion and control using C++ for 115lb competition robot
- Integrating computer vision localization with on-the-fly A* pathfinding to power fully autonomous scoring
- World Comp. Autonomous Award & Division Champions. **Rated #1 globally for autonomous performance**

Research Assistant — Jan 2026 – Present

Georgia Tech Engineering Space Policy Lab — *Atlanta, GA*

- Modeling the downrange risk of space launches to inform policy decisions on launch safety and risk mitigation
- Scraping FAA NOTAMs to build a dataset of launch attempts; Plotting risk contours using QGIS

Backend Software Engineering Intern — May 2026 – Aug 2026

Ramp — *New York, NY*

- Will support development of Procure-to-Pay (P2P) platform automation for enterprise clients

Software Engineering Intern — Aug 2024 – May 2025

Viasat Inc. — *Duluth, GA*

- Developed C/C++ firmware and Typescript frontend for prototype three-axis satellite antennas
- Built algorithms for analyzing signal quality and atmospheric modeling; integrated CI pipelines for validation

UX Assistant — Aug 2025 – Present

Georgia Tech PACE High Performance Computing Center — *Atlanta, GA*

- Supporting roughly 5,000 researchers across high-performance computing systems with incident triage

Collegiate Activites

Guidance, Navigation and Control Team — Aug 2025 – Present

Propulsive Landers @ Georgia Tech — *Atlanta, GA*

- Applying Klaman Filters and nonlinear MPC in Rust for 6 DOF control of rocket lander and UAV prototypes
- Implementing lossless convexification for faster on-device fuel-optimal flight trajectory replanning in Rust
- Prototyping with Monoprop UAV (drone); Aiming for 50m hop with 1000N rocket by EOY 2026

RoboCup: Autonomous Software Team — Aug 2025 – Present

RoboJackets @ Georgia Tech — *Atlanta, GA*

- Building multi-agent pathfinding, obstacle avoidance and task allocation in C++ for 6v6 robot soccer

Projects & Research Publications

Evaluating Numerical Solving Methods for Optimal Powered Descent — Mar 2026

AIAA Region II Student Conference — *Columbia, SC*

- Compared ZOH discretization and psuedospectral paramterization to plan 3 DOF trajectories for a rocket lander
- Minimized error from optimal solution at coarse solver resolutions for fast on-the-fly replanning (sub 10ms)

Holonomic Auto-Alignment using CV Localization | *C++, Java, ArUco* — Aug 2023 – May 2025

- Applied second-order kinematics and jerk limiting to replicate S-curve profile for discretized holonomic drive
- Fused CV estimates from fiducials with filtered odometry using a state-space model for real-time robot location
- Allowed competition robot to align to any scoring position with < 1cm accuracy regardless of starting state

Technical Skills

Computer Languages: Python, Java, C, C++, TypeScript, Rust

Frameworks & Libraries: TensorFlow, YOLO, Jupyter, ROS, Matplotlib, OpenCV, ArUco, React